RQ Answers

omitted

2024-04-05

Data overview

```
glimpse(all_data)
```

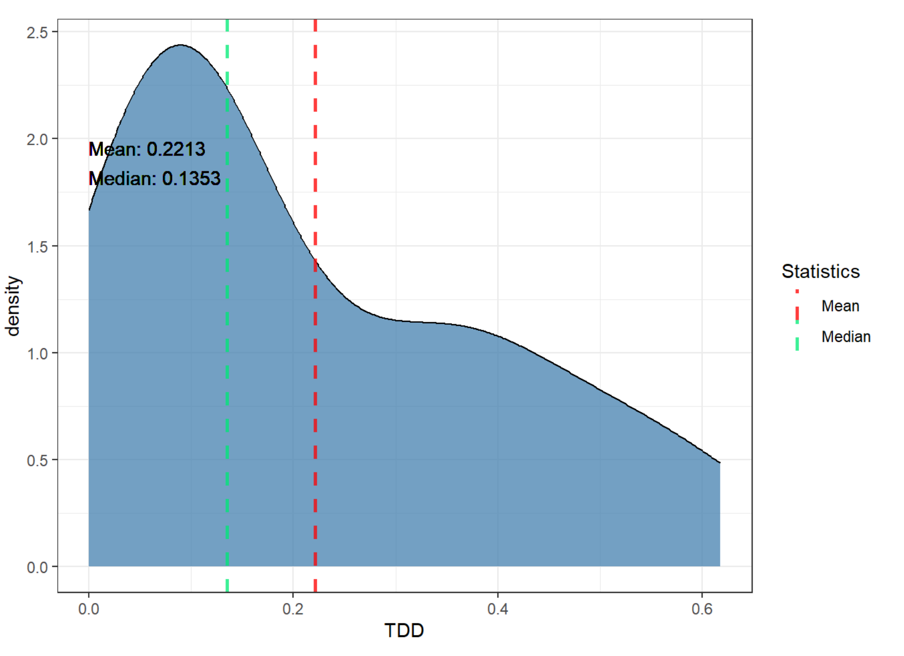
```
## Rows: 391
## Columns: 11
## $ repo              <chr> "maven-surefire", "maven-surefire", "maven-surefir…
## $ pr_number         <int> 396, 396, 396, 396, 396, 396, 396, 396, 396, 396, …
## $ rule              <chr> "java:S1133", "java:S1135", "java:S1133", "java:S1…
## $ severity          <chr> "INFO", "INFO", "INFO", "INFO", "INFO", "MAJOR", "…
## $ file              <chr> "maven-surefire-common/src/main/java/org/apache/ma…
## $ type              <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM…
## $ status            <chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O…
## $ debt              <int> 10, 0, 10, 0, 10, 5, 2, 2, 5, 20, 0, 20, 13, 5, 38…
## $ ncloc_affected_file <int> 3123, 3123, 3123, 3123, 3123, 3123, 3123, 3123, 31…
## $ complexity        <int> 599, 599, 599, 599, 599, 599, 599, 599, 599, 599, …
## $ origin            <chr> "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTING", "P…
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the da
ta has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but th
e data has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

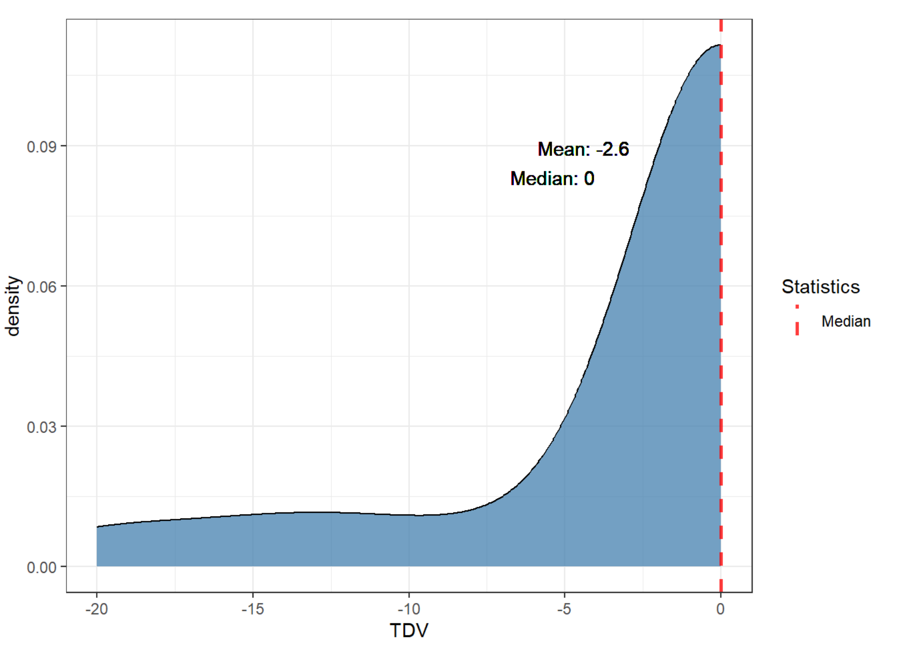


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the da
ta has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but th
e data has 20 rows.
## ℹ Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 1 × 4
##   repo           tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>                        <dbl>             <dbl>                 <dbl>
## 1 maven-surefire                  20                80                     0
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 × 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##                      <dbl>                  <dbl>                      <dbl>
## 1                       20                     80                          0
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 1 × 3
##   repo           preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>                                  <dbl>                             <dbl>
## 1 maven-surefire                            80                                20
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##                           <dbl>                                  <dbl>
## 1                            80                                     20
```

Percentages for new TDV by repo

```
## # A tibble: 1 × 3
## # Groups:   repo [1]
##   repo           new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>                          <dbl>                     <dbl>
## 1 maven-surefire                   100                         0
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 × 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##                        <dbl>                          <dbl>
## 1                        100                              0
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (../issues_characterization.Rmd), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the $PositionScore$ metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^{k} Position_j$$

given a rule i, its $PositionScore_i$ will be the sum of its position across all k repositiries.After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the PositionScore, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 5 × 2
## # Groups:   rank [5]
##    rank `maven-surefire`
##   <int> <chr>
## 1     1 java:S3740  -  3
## 2     2 java:S1192  -  2
## 3     3 java:S107  -  1
## 4     4 java:S1874  -  1
## 5     5 java:S3776  -  1
```

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

```
## # A tibble: 5 × 2
##   rule       position_score
##   <chr>               <dbl>
## 1 java:S3740            122
## 2 java:S1192            123
## 3 java:S107             124
## 4 java:S1874            125
## 5 java:S3776            126
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 × 2
## # Groups:   rank [10]
##     rank `maven-surefire`
##    <int> <chr>
##  1     1 java:S3776  -  37
##  2     2 java:S1874  -  35
##  3     3 java:S1133  -  34
##  4     4 java:S1135  -  33
##  5     5 java:S112  -  26
##  6     6 java:S1123  -  25
##  7     7 java:S1192  -  22
##  8     8 java:S1186  -  21
##  9     9 java:S1117  -  16
## 10    10 java:S3077  -  15
```

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

```
## # A tibble: 10 × 2
##    rule       position_score
##    <chr>               <dbl>
##  1 java:S3776            122
##  2 java:S1874            123
##  3 java:S1133            124
##  4 java:S1135            125
##  5 java:S112             126
##  6 java:S1123            127
##  7 java:S1192            128
##  8 java:S1186            129
##  9 java:S1117            130
## 10 java:S3077            131
```

Outliers

Higher TDD

```
## # A tibble: 20 × 5
##    repo           pr_number total_debt total_ncloc    tdd
##    <chr>              <int>      <int>       <int>  <dbl>
##  1 maven-surefire       548        139         225 0.618
##  2 maven-surefire       435         85         155 0.548
##  3 maven-surefire       669        200         419 0.477
##  4 maven-surefire       641        145         337 0.430
##  5 maven-surefire       454        362        1024 0.354
##  6 maven-surefire       545        100         293 0.341
##  7 maven-surefire       636        150         440 0.341
##  8 maven-surefire       619       1170        4253 0.275
##  9 maven-surefire       396        498        3334 0.149
## 10 maven-surefire       502        418        3088 0.135
## 11 maven-surefire       508        418        3089 0.135
## 12 maven-surefire       444        133        1017 0.131
## 13 maven-surefire       596         40         527 0.0759
## 14 maven-surefire       555         40         528 0.0758
## 15 maven-surefire       527         20         281 0.0712
## 16 maven-surefire       668         11         175 0.0629
## 17 maven-surefire       490         30         497 0.0604
## 18 maven-surefire       564         18         313 0.0575
## 19 maven-surefire       608         18         325 0.0554
## 20 maven-surefire       643          5         161 0.0311
```

Higher TDV

```
## # A tibble: 20 × 3
##    pr_number repo             tdv
##        <int> <chr>          <int>
##  1       444 maven-surefire     0
##  2       490 maven-surefire     0
##  3       502 maven-surefire     0
##  4       508 maven-surefire     0
##  5       527 maven-surefire     0
##  6       545 maven-surefire     0
##  7       548 maven-surefire     0
##  8       555 maven-surefire     0
##  9       564 maven-surefire     0
## 10       596 maven-surefire     0
## 11       619 maven-surefire     0
## 12       636 maven-surefire     0
## 13       641 maven-surefire     0
## 14       643 maven-surefire     0
## 15       668 maven-surefire     0
## 16       669 maven-surefire     0
## 17       608 maven-surefire    -6
## 18       454 maven-surefire   -11
## 19       435 maven-surefire   -15
## 20       396 maven-surefire   -20
```

Lower TDV

```
## # A tibble: 20 × 3
##    pr_number repo             tdv
##        <int> <chr>          <int>
##  1       396 maven-surefire   -20
##  2       435 maven-surefire   -15
##  3       454 maven-surefire   -11
##  4       608 maven-surefire    -6
##  5       444 maven-surefire     0
##  6       490 maven-surefire     0
##  7       502 maven-surefire     0
##  8       508 maven-surefire     0
##  9       527 maven-surefire     0
## 10       545 maven-surefire     0
## 11       548 maven-surefire     0
## 12       555 maven-surefire     0
## 13       564 maven-surefire     0
## 14       596 maven-surefire     0
## 15       619 maven-surefire     0
## 16       636 maven-surefire     0
## 17       641 maven-surefire     0
## 18       643 maven-surefire     0
## 19       668 maven-surefire     0
## 20       669 maven-surefire     0
```

Extra

Number of PRs with pre-existing TD

```
##    n
## 1 20
```

Number of java:S1192 issues that affect test code

```
##   n
## 1 0
```

Number of java:S1192 issues that affect not test code

```
##    n
## 1 24
```

Summary NCLOC

```
##      ncloc
##  Min.   : 155.0
##  1st Qu.: 290.0
##  Median : 429.5
##  Mean   :1021.2
##  3rd Qu.:1018.8
##  Max.   :4197.0
```